

Objective: To process the data and create a visualization using the pre-processed dataset.

Dataset:

- The link also has a folder which contains PR and GHI data, organized into directories based on the year-month and parameter.
- PR (Performance Ratio) – This parameter is used to track the daily performance of the PV plant. A high value indicates that the plant is performing well and there are no issues. GHI (Global Horizontal Irradiance) – This parameter tracks the total irradiation for a particular day. A high value indicates a sunny day.
- Below is an example of the folder structure:

- ❖ PR/

- ✓ 2023-01/
- ✓ 2023-01-01_PR.csv
- ✓ 2023-01-06_PR.csv

- ❖ GHI/

- ✓ 2023-01/
- ✓ 2023-01-01_GHI.csv
- ✓ 2023-01-06_GHI.csv

Data Processing:

- Generate a single CSV file containing all the data from both the PR and GHI folders. The new CSV file should contain 3 columns: Date, GHI, PR.
- Important Notes:
 - ✚ The data has to be collated into a single file. The file should contain 982 rows.
 - ✚ -Create a single function to preprocess the data. Make sure you organize your code and it is readable.

Data Visualization:

- Once the data has been processed, the below graph must be generated using this processed data:
- Important Notes:
 - ✚ Create a single function to generate the graph. Make sure you organize your code and it is readable.
 - ✚ The red line on the graph represents the 30-d moving average of the PR (Performance Evolution) whereas the position of the scatter points depict the PR value of that day and the colour shade depicts the GHI value.

✚ The dark green line represents the budget line. The value begins from 73.9 and should reduce by 0.8% every year (Do not hardcode the values). As you can see, the values are:

- 73.9 for the first year (July 2019 to June 2020)
- 73.1 for the second year (July 2020 to June 2021)
- 72.3 for the third year (July 2021 to June 2022)
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➤ This should happen dynamically via code. The points for the scatter plot are color-coded (as per the legend). If the GHI [Daily Irradiation] is:

- Less than 2: Navy blue
- 2-4: Light blue
- 4-6: Orange
- >6: Brown

➤ The points above Target Budget PR represent the number of PR points above the Budget PR for that particular year.

➤ The bottom right section of the graph simply shows the average PR for the last 7 days, the last 30 days, the last 60 days, and so on.

